Assignment 2: Offline model evaluation using Deepchecks (2 points)

Model Evaluation Suite

The suite is composed of the following checks: Model Inference Time, Train Test Performance.

Each check may contain conditions (which will result in pass ✓ / fail ✖ / warning ! / error ?!) as well as other outputs such as plots or tables.

Suites, checks and conditions can all be modified. Read more about custom suites.

Didn't Pass

Status	Check	Condition	More Info
✖	Train Test Performance	Train-Test scores relative degradation is less than 0.2	2 scores failed. Found max degradation of 47.8% for metric neg_mae

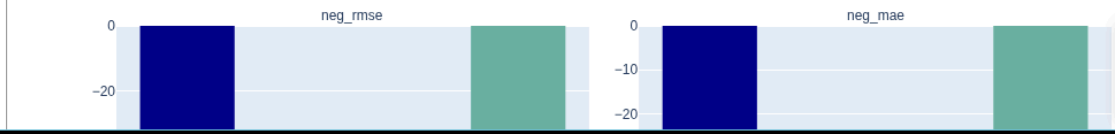
Train Test Performance

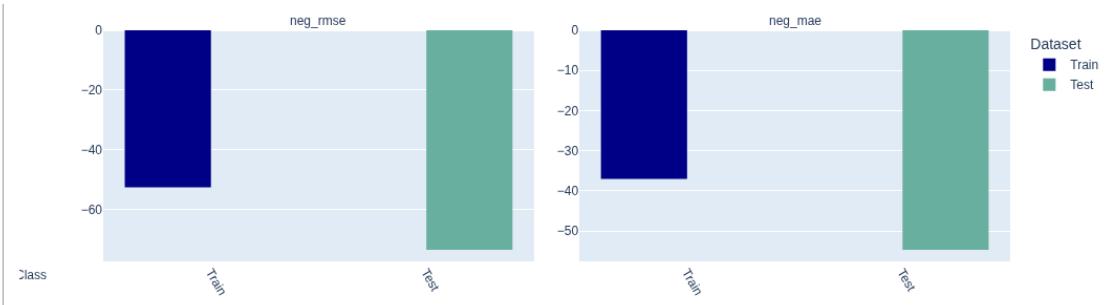
Summarize given model performance on the train and test datasets based on selected scorers. Read More...

Conditions Summary

Status	Condition	More Info
✖	Train-Test scores relative degradation is less than 0.2	2 scores failed. Found max degradation of 47.8% for metric neg_mae

Additional Outputs





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Passed

Status	Check	Condition	More Info
✓	Model Inference Time - Train Dataset	Average model inference time for one sample is less than 0.1	Found average inference time (seconds): 1.83e-06
✓	Model Inference Time - Test Dataset	Average model inference time for one sample is less than 0.1	Found average inference time (seconds): 3.67e-06

Model Inference Time - Train Dataset

Measure model average inference time (in seconds) per sample. Read More...

Conditions Summary

Conditions Summary

Status	Condition	More Info
✓	Average model inference time for one sample is less than 0.1	Found average inference time (seconds): 1.83e-06

Additional Outputs

Average model inference time for one sample (in seconds): 1.83e-06

Note - data sampling: Data is sampled from the original dataset, running on 1000 samples out of 10718. Sample size can be controlled with the "n_samples" parameter.

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Model Inference Time - Test Dataset

Measure model average inference time (in seconds) per sample. Read More...

Conditions Summary

Status	Condition	More Info
✓	Average model inference time for one sample is less than 0.1	Found average inference time (seconds): 3.67e-06

Additional Outputs

Average model inference time for one sample (in seconds): 3.67e-06

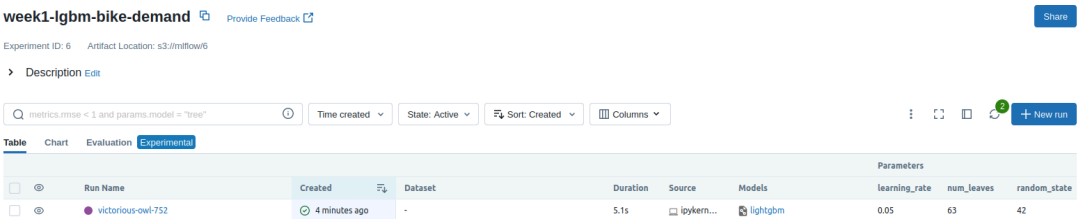
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Assignment 3: Tracking model training in MLflow (2 points)

Screenshots to be submitted for Assignment 3

To get the points from Assignment 3, please submit the following screenshots:

1. The logs of your MLflow run. Please include the parameters of the model in your screenshot.



2. The details of the MLflow run, including the uploaded Deepchecks evaluation result file;

week1-lgbm-bike-demand ›

victorious-owl-752

Run ID: 0a4bbdf5624f4b01b4734a8c8fa1a585 Date: 2026-11-01 19:19:51 Source: ipykernel_launcher.py User: user Duration: 5.1s

Status: FINISHED Lifecycle Stage: active

- Description Edit
- Datasets
- Parameters (3)
- Metrics
- Tags
- Artifacts

Week1LgbmBikeDemand
- MLmodel
- conda.yaml
- model.pkl
- python_env.yaml
- requirements.txt
- evaluation_result.html

Full Path:s3://mlflow/6/0a4bbdf5624f4b01b4734a8c8fa1a585/artifacts/evaluation_result.html
Size: 7.36MB

▾ Model Evaluation Suite

Model Evaluation Suite

The suite is composed of the following checks: Train Test Performance, Model Inference Time.

Each check may contain conditions (which will result in pass ✓ / fail ✖ / warning ! / error ?!) as well as other outputs such as plots or tables.

Suites, checks and conditions can all be modified. Read more about custom suites.

▾ Model Evaluation Suite

Model Evaluation Suite

The suite is composed of the following checks: Train Test Performance, Model Inference Time.

Each check may contain conditions (which will result in pass ✓ / fail ✖ / warning ! / error ?!) as well as other outputs such as plots or tables.

Suites, checks and conditions can all be modified. Read more about custom suites.

▾ Didn't Pass

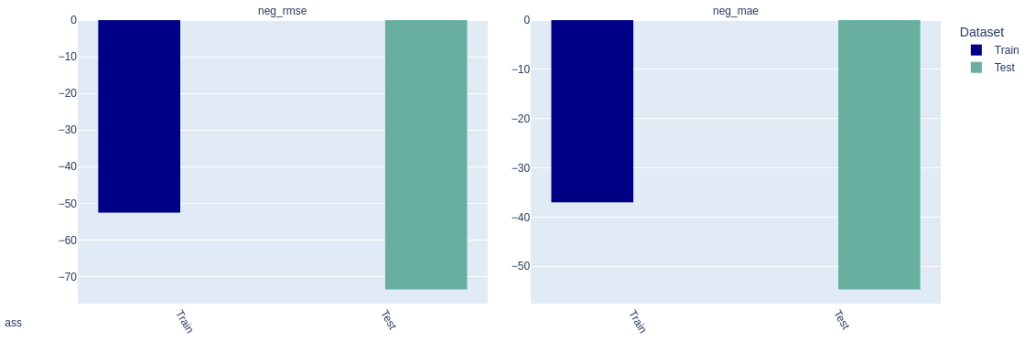
Status	Check	Condition	More Info
✖	Train Test Performance	Train-Test scores relative degradation is less than 0.2	2 scores failed. Found max degradation of 47.8% for metric neg_mae

Train Test Performance

Summarize given model performance on the train and test datasets based on selected scorers. Read More...

Conditions Summary

Status	Condition	More Info
✖	Train-Test scores relative degradation is less than 0.2	2 scores failed. Found max degradation of 47.8% for metric neg_mae



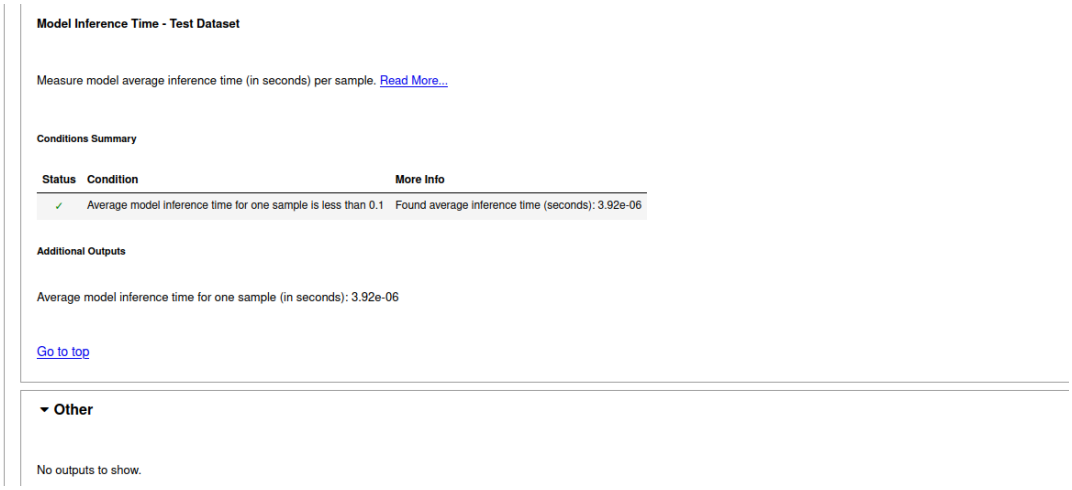
ass

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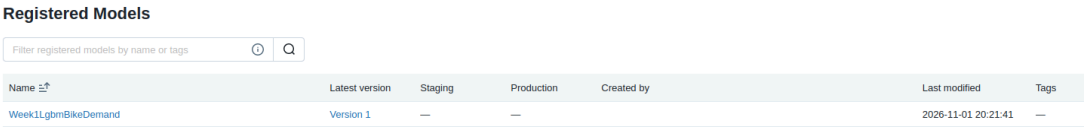
▾ Passed

Status	Check	Condition	More Info
✓	Model Inference Time - Train Dataset	Average model inference time for one sample is less than 0.1	Found average inference time (seconds): 1.87e-06





3. The registered model.



Assignment 4: Evaluate the trained model against another model (2 points)

Screenshots to be submitted for Assignment 4

The details of the MLflow run including uploaded Deepchecks model comparison result file.

honorable-ape-138

Run ID: 31d10eb257514c9589bdabd997979baf Date: 2026-11-02 15:25:23 Source: ipykernel_launcher.py User: user Duration: 4.8s Status: UNFINISHED Lifecycle Stage:

› Description Edit

› Datasets

⌄ Parameters (3)

Name	Value
learning_rate	0.05
num_leaves	63
random_state	42

› Metrics

› Tags

⌄ Artifacts

- model
  - MLmodel
  - conda.yaml
  - model.pkl
  - python_env.yaml
  - requirements.txt
- evaluation_result.html
- model_comparison.html

Full Path:s3://mlflow/6/31d10eb257514c95...
Size: 7.32MB

Multi Model Performance Report

Summarize performance scores for multiple models on test datasets. Read More...

Additional Outputs

